CLOSED POINTS WITH FINITE SEPARABLE RESIDUE FIELD ARE DENSE ON A SMOOTH SCHEME OVER A FIELD

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the closedness of the constructed point; no step is referred to the reader.

Editorial note. The counted spans carry 5 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-smooth-separable-closed-points-dense is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Smooth schemes are etale locally standard smooth, providing the etale map from an open of affine space. Morphisms, Lemma morphisms-lemma-smooth-etale-over-affine-space, invoked at proof lines 4677-4678. Supplies the etale morphism pi used in the argument.
- Etale morphisms are open and their residue field extensions over a field are finite separable. Morphisms, Lemmas morphisms-lemma-etale-open and morphisms-lemma-etale-over-field, invoked at proof lines 4679-4686. Supplies openness of pi and finite separability of kappa(x)/kappa(w).
- Existence of a closed point with finite separable residue field in a nonempty open of affine space. Same chapter, Lemma lemma-affine-space-over-field, invoked at proof lines 4681-4683. Supplies the point w in the image of pi.
- A point whose residue field is algebraic over the base field is closed. Morphisms, Lemma morphisms-lemma-algebraic-residue-field-extension-closed-point-fibre, invoked at proof lines 4687-4690. Supplies that the lifted point x is closed in X.

Source reference. Original lemma 055T: sources/stacks-varieties/pinned-original-source.tex, source lines 4643–4643. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01TE: sources/stacks-chow/tags.txt, source lines 2371–2371. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 03WT: sources/stacks-chow/tags.txt, source lines 5124–5124. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02GL: sources/stacks-chow/tags.txt, source lines 3232–3232. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 054L: sources/stacks-chow/tags.txt, source lines 6644–6644. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. If X is smooth over* $\mathrm{Spec}(k)$ *then the set*

$$\{x \in X \text{ closed such that } k \subset \kappa(x) \text{ is finite separable}\}$$

is dense in X.

Proof. It suffices to show that given a nonempty smooth X over k there exists at least one closed point whose residue field is finite separable over k. To see this, choose a diagram

$$X \leftarrow U \xrightarrow{\pi} \mathbf{A}^d_k$$

with π étale, see Morphisms, Lemma 054L. The morphism $\pi : U \to \mathbf{A}^d_k$ is open, see Morphisms, Lemma 03WT. By Lemma 055T we may choose a closed point $w \in \pi(U)$ whose residue field is finite separable over k. Pick any $x \in U$ with $\pi(x) = w$. By Morphisms, Lemma 02GL the field extension $\kappa(x)/\kappa(w)$ is finite separable. Hence $\kappa(x)/k$ is finite separable. The point x is a closed point of X by Morphisms, Lemma 01TE. □